OLIVER ZHANG

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EDUCATION

The University of Texas at Austin May 2028
Turing Scholar Honors | B.S. Computer Science | GPA: 3.61 / 4.00
Coursework: Operating Systems, Artificial Intelligence, Data Structures & Algorithms, Computer Organization & Architecture, Probability & Statistics, Linear Algebra, Generalized Linear Models

EXPERIENCE

tradeInsight.ai – Software Engineer Intern, socialInsight September 2026 – November 2026
AI market-intelligence startup; socialInsight turns online discussion into company-level data

- Evaluating a third-party social-data source for the socialInsight data pipeline on its coverage, data quality and research value
- Designing how data from the new source is collected, filtered and integrated into the startup's existing social-data pipeline

IBM – Software Engineering Intern; Research Triangle Park, NC May 2026 – August 2026
AI operations agent: anomaly detection and root-cause triage for IBM Z mainframes

- Built the agent with one other engineer, from an empty repo to running on customer IBM Z mainframe logs in 10 weeks
- Built the Gemini multi-agent orchestrator on Vertex AI (ADK), routing each question to an analytics agent or a research agent
- Scoped and built the integration between the Gemini data-analytics agent and the orchestrator that the platform did not support

Tianjin University – Research Intern; Tianjin, China May 2025 – August 2025

- Delivered a real-time computer-vision tour guide pairing on-device detection with an LLM, at 30 FPS and sub-2s latency on CPU

PROJECTS

KRONOS: Regime-Aware Quant Research Platform · Python (NumPy/SciPy), HMM, Kalman, GARCH, HRP 2026 – Present

- Developed a regime-aware quant research platform from scratch in Python, hand-coding HMM, Kalman, GARCH and risk parity
- Ran a return-attribution self-audit that exposed an inverted drawdown throttle, a sign bug 30 green verification gates had missed

Custos Code: Auditing What Coding Agents Report · Python, LLM agents, CLI September 2026

- Led a 3-person hackathon team building a tool that checks an AI coding agent's final report against the log of what it actually did
- Designed the core engine, which labels each of the agent's claims confirmed, contradicted or unwitnessed against its own action log

LLM Memory Layer (inf-mem) · Python, FastAPI, ONNX, SQLite, Chrome Extension 2025 – 2026

- Gave ChatGPT and Claude persistent memory: a Chrome extension and FastAPI backend, Llama 3.1 8B extraction, BGE retrieval

AI Charity Diligence Pipeline · Python, LLM tiering, IRS 990 data 2026 – Present

- Designed a screening pipeline for a student philanthropy fund, routing each task to the cheapest capable executor: rules, then LLMs

x86-64 Optimizing Compiler (Course Project) · C, x86-64 Assembly Spring 2026

- Wrote an x86-64 optimizing compiler in C for a statically typed language; generated code ran 60× faster than the interpreter
- Optimized register allocation, stack-frame management, and calling-convention compliance, verified by a correctness test suite

High-Performance Matrix Multiplication (Course Project) · C, AVX2, pthreads, Cachegrind Spring 2026

- Implemented 9 algorithmic variants (naive, cache-blocked, recursive, SIMD, multithreaded) in C, peaking at 56.1 GFLOPS

RESEARCH PUBLICATIONS

IEEE IPCCC: Evaluating the Carbon Impact of Large Language Models at the Inference Stage March 2023 – November 2023
Coauthor | Texas Tech University

- Profiled 5 LLMs across CPU and GPU with a Python power framework; a mid-size model matched quality at 73% less energy

IARIA GREEN: Cost and Carbon Reduction for Microsoft Azure Virtual Machines Using Workload Analysis September 2023
Coauthor

- Analyzed workload data from 2.6M+ Microsoft Azure VMs for cost and carbon reduction, modeling cost from core/memory specs

HONORS & AWARDS

Citadel Securities Engineering Challenge at UT Austin: 3rd Place (January 2026)

SKILLS

Languages: C, C++, Java, Python, JavaScript, SQL, x86-64 Assembly, Bash
Core CS: Data Structures & Algorithms, object-oriented software design, operating systems, compilers, concurrency (pthreads)
Systems & Scale: performance optimization, distributed systems, scalability, debugging & profiling (GDB, Valgrind)
Data & Backend: databases (SQLite, BigQuery), REST/WebSocket APIs, client–server design, Docker, Linux/Unix, Git
Machine Learning: PyTorch, TensorFlow/TFLite, OpenCV, NumPy, Pandas, LangChain